

# 7.1 An Introduction to A Level Organic Chemistry (A Level Only)

## Question Paper

|            |                                                                 |
|------------|-----------------------------------------------------------------|
| Course     | CIE A Level Chemistry                                           |
| Section    | 7. Organic Chemistry (A Level Only)                             |
| Topic      | 7.1 An Introduction to A Level Organic Chemistry (A Level Only) |
| Difficulty | Medium                                                          |

**Time allowed:** 50

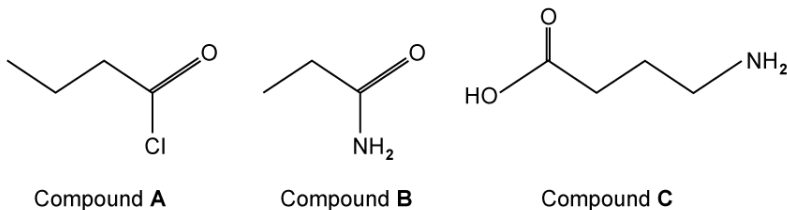
**Score:** /41

**Percentage:** /100

### Question 1a

Organic compounds can be represented using skeletal formula.

Give the IUPAC names of the compounds **A**, **B** and **C**.



[3 marks]

### Question 1b

Compound **A** can undergo hydrolysis to form a carboxylic acid.



Explain why this is an example of an addition-elimination reaction.

[2 marks]

### Question 1c

When heated, compound **A** will also react with phenol in the presence of a base to produce phenyl butanoate.

Draw the displayed formula of phenyl butanoate.

[2 marks]

**Question 1d**

State whether any of the compounds, **A**, **B** or **C**, can form optical isomers.

Explain your answer.

[2 marks]

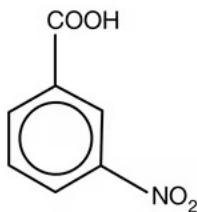
**Question 2a**

Benzene can exist as  $C_6H_6$  or be substituted in a number of different ways to form aromatic compounds.

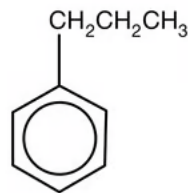
State the IUPAC name for compounds **W**, **X** and **Y**.



Compound **W**



Compound **X**

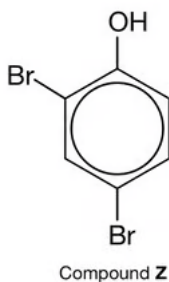


Compound **Y**

[3 marks]

**Question 2b**

State the IUPAC name for compound Z.



[1 mark]

**Question 2c**

State the bond angle that is present in the benzene molecule.

[1 mark]

**Question 2d**

In benzene, each of the carbon atoms is  $sp^2$  hybridised.

i)

Explain how neighbouring carbon atoms form  $\sigma$  bonds.

[1]

ii)

Neighbouring carbon atoms also form  $\pi$  bonds. Explain how these  $\pi$  bonds are formed.

[1]

[1 mark]

### Question 3a

This question is about benzene.

i)

Explain why the benzene molecule is planar.

[2]

ii)

Explain why all of the carbon-carbon bonds in benzene are the same length.

[1]

[3 marks]

### Question 3b

Benzene can be nitrated by warming it with a mixture of concentrated sulfuric acid and concentrated nitric acid.

The mechanism for the reaction is shown in Fig. 3.1.

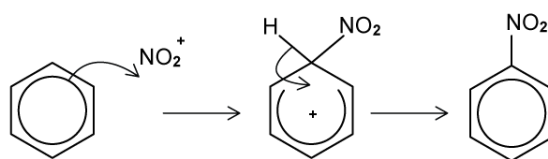


Fig. 3.1

Name the type of mechanism of this nitration reaction. Explain your answer.

[3 marks]

### Question 3c

Benzene can be used to synthesise (4-aminophenyl)ethanoic acid which is a drug that can be used to treat chronic inflammation of the intestines.

i)

Draw the structural formula of (4-aminophenyl)ethanoic acid.

[1]

ii)

One of the intermediate products made during the synthesis of (4-aminophenyl)ethanoic acid is the molecule shown in Fig. 3.2.



**Fig. 3.2**

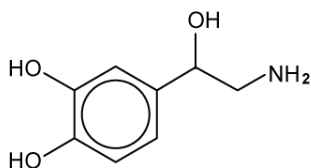
Give the IUPAC name of the molecule shown in Fig. 3.2.

[1]

[2 marks]

### Question 3d

Noradrenaline, shown in Fig. 3.3, is a hormone and neurotransmitter which contains a benzene ring. It is released during stress to stimulate the heart and increase blood pressure.



**Fig. 3.3**

Other than the arene functional group, state the **name** of **three** functional groups in the noradrenaline molecule.

[3 marks]

**Question 4a**

Leucine is an optically active amino acid with the molecular formula  $(\text{CH}_3)_2\text{CHCH}_2\text{CH}(\text{NH}_2)\text{COOH}$ .

Draw the displayed formula of leucine.

[1 mark]

**Question 4b**

Leucine exists as stereoisomers.

i)

Explain the meaning of the term **stereoisomers**.

[2]

ii)

Explain how you can distinguish between the isomers.

[2]

[4 marks]

### Question 4c

The two optical isomers of leucine taste different. One isomer has a bitter taste, the other is sweet.

Draw the three-dimensional arrangement of the two optical isomers.

[2 marks]

### Question 4d

Give the IUPAC name of leucine.

[1 mark]

### Question 5a

Propranolol is an optically active drug that is used to treat hypertension. Its structure is shown in Fig. 5.1.

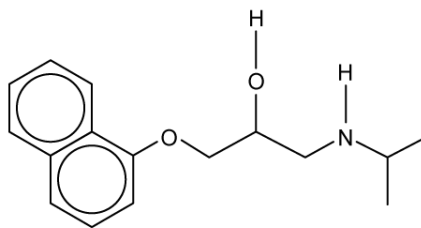


Fig. 5.1

- i)  
Give the molecular formula of propranolol.

[1]

- ii)  
Circle the chiral centre of propranolol in Fig 5.1.

[1]

[2 marks]



**Question 5b**

Explain why a racemic mixture of propranolol has no effect on plane polarised light.

[3 marks]

**Question 5c**

Explain why it is important to separate a racemic mixture of propranolol into pure single enantiomers.

[1 mark]

**Question 5d**

Chiral catalysts are often used in the manufacturing process of drugs.

i)

What is the purpose of a chiral catalyst?

[1]

ii)

Give **one** advantage and **one** disadvantage of using a chiral catalyst.

[2]

[1 mark]

## Model Answers: Medium

1a

a) The IUPAC names of the compounds **A**, **B** and **C** are:

- Compound **A** – butanoyl chloride; [1 mark]
- Compound **B** – propanamide; [1 mark]
- Compound **C** – 4–aminobutanoic acid; [1 mark]

### [Total:3 marks]

- Naming compound **A**:
  - Determine the primary functional group: acyl chloride,  $-\text{COCl}$ 
    - Suffix of name –oyl chloride
  - Determine the primary chain: 4 carbons with only single bonds
    - Stem of name butan–
  - The acyl chloride can only go on an end carbon so it does not need to be numbered
  - This gives the name butanoyl chloride
- Naming compound **B**
  - Determine the primary functional group: amide,  $-\text{CONH}_2$ 
    - Suffix of name –amide
  - Determine the primary chain: 3 carbons with only single bonds
    - Stem of name propan–
  - The amide can only go on an end carbon so it does not need to be numbered
  - This gives the name propanamide
- Naming compound **C**
  - Determine the functional groups: amino,  $-\text{NH}_2$  and carboxylic acid,  $-\text{COOH}$ 
    - Prefix of name, amino–
    - Suffix of name –oic acid
  - Determine the primary chain: 3 carbons with only single bonds
    - Stem of name propan–
  - The carboxylic acid can only go on an end carbon so it does not need to be numbered but the position of the amino group needs to be given in relation to the carboxylic acid

- The carbon of the  $\text{-COOH}$  is given position 1, so the  $\text{-NH}_2$  group is on carbon 4
- This gives the name 4-aminobutanoic acid

1b

b) This is an example of an addition-elimination reaction because:

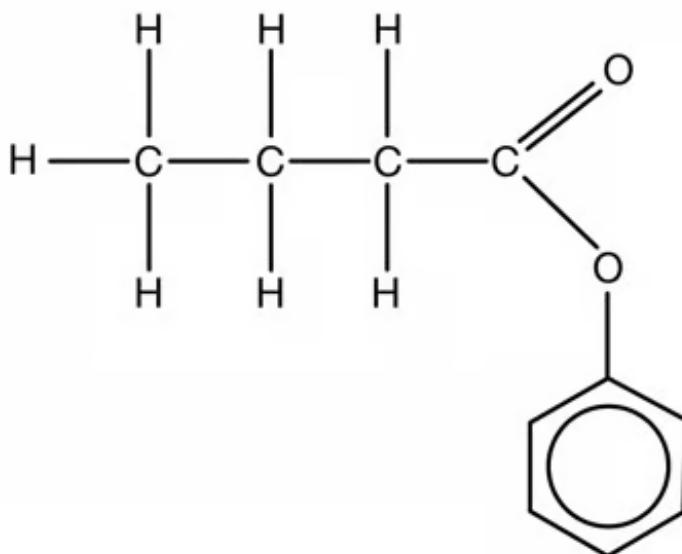
- Water adds across the  $\text{C=O}$  bond; [1 mark]
- $\text{HCl}$  is eliminated; [1 mark]

**[Total: 2 marks]**

- You need to understand the terminology associated with various mechanisms:
  - free-radical substitution
  - electrophilic addition
  - nucleophilic substitution
  - nucleophilic addition
  - electrophilic substitution
  - addition-elimination
- An addition-elimination reaction is a reaction in which 2 molecules join together with the loss of a small molecule, in this case,  $\text{HCl}$

1c

c) The displayed formula of phenyl butanoate is:



- Correct ester link  $\text{-COO-}$ ; [1 mark]
- Rest of structure correct; [1 mark]

**[Total: 2 marks]**

- 'Butan-' tells you that the carbon chain has 4 carbon atoms with single bonds between each carbon atom
- '-oate' tells you that it is an ester, so draw the  $\text{-COO-}$  group on one of the end carbon atoms, remembering that there is a double bond to one oxygen and a single bond to the other
- The first part of the name of an ester comes from the alcohol
  - Phenyl indicates that phenol was used
  - The H in the OH group of phenol is lost, leaving the O
  - This is the O of the C-O of the ester link, so you need to draw a benzene ring from this O

1d

d) Stating whether any of the compounds, **A**, **B** or **C**, can form optical isomers and why:

- None of the compounds form optical isomers; [1 mark]
- Because none have a chiral centre; [1 mark]

**[Total: 2 marks]**

- Compounds with a chiral centre exist as two optical isomers which are also known as enantiomers
- A carbon atom that has four different atoms or groups of atoms attached to it is called a chiral carbon or chiral centre
- None of these compounds has a carbon atom which is attached to four different atoms or groups of atoms

2a

a) The IUPAC names for compounds **W**, **X** and **Y** are:

- **W** = 4-chloromethylbenzene; [1 mark]
- **X** = 3-nitrobenzoic acid; [1 mark]
- **Y** = propylbenzene; [1 mark]

**[Total: 3 marks]**

- The benzene ring is the main functional group in compounds **W** and **Y** so the suffix is -benzene
- In **W**, the prefixes chloro- and methyl- are needed
  - The methyl group assumes position 1 and the chloro- group is given a position in relation to this i.e. 4
- In **Y** the prefix propyl- is used to represent the  $-\text{CH}_2\text{CH}_2\text{CH}_3$  group
- For compound **Z** the benzene ring has the  $-\text{COOH}$  group attached, so its suffix is benzoic acid
  - It also has the  $\text{NO}_2$  group attached, so the prefix nitro- is needed, and its position is given in relation to the  $\text{COOH}$  group

2b

b) The IUPAC name for compound **Z** is:

- 2,4-dibromophenol; [1 mark]

**[Total: 1 mark]**

- This benzene ring contains an  $-\text{OH}$  group so the main stem is phenol
- Starting from the  $\text{OH}$  group (on carbon number 1 as this is the dominant functional group) and counting anticlockwise, there are two bromine atoms on carbons 2 and 4; this is the lowest numbered version as in this case counting clockwise would result in numbers of 4 and 6
- So the name is given as 2,4-dibromophenol

2c

c) The bond angle in benzene is:

- $120^\circ$ ; [1 mark]

**[Total: 1 mark]**

- As each carbon atom is bonded to three other atoms the bond angle will be  $120^\circ$  and the shape will be trigonal planar
- The three areas of electron density arrange themselves as far apart from each other as possible, hence the angle achieved, and there are no lone pairs on the

carbons to distort this angle

2d

d)

i) The  $sp^2$  orbitals form  $\sigma$  bonds by:

- Overlapping of  $sp^2$  orbitals; [1 mark]

ii)  $\pi$  bonds are formed by:

- Overlapping of p orbitals; [1 mark]

**[Total: 2 marks]**

- 3 of the orbitals in each carbon atom are hybridised
- The hybridised orbital is made from 1 s orbital and 2 p orbitals, hence they are called  $sp^2$  hybrids

- Starting from the OH group (on carbon number 1 as this is the dominant functional group) and counting anticlockwise, there are two bromine atoms on carbons 2 and 4; this is the lowest numbered version as in this case counting clockwise would result in numbers of 4 and 6
- So the name is given as 2,4-dibromophenol

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**[Total: 2 marks]**

- 3 of the orbitals in each carbon atom are hybridised
- The hybridised orbital is made from 1 s orbital and 2 p orbitals, hence they are called  $sp^2$  hybrids
- Each carbon will overlap its  $sp^2$  hybridised orbital with the  $sp^2$  hybridised orbital of 2 neighbouring carbon atoms and an s orbital of a hydrogen atom
  - These are sigma,  $\sigma$ , bonds and they arrange themselves at  $120^\circ$  to each other in one plane

- Each carbon atom has a fourth electron in a p orbital at right angles to the  $sp^2$  orbitals, so above and below the plane of the rest of the molecule
  - The p orbitals overlap forming a  $\pi$  bond
  - It is these electrons in the p orbitals which become delocalised and can spread around the ring

3a

a)

i) Benzene is planar because:

- The carbons are  $sp^2$  hybridised / trigonal planar / bonded at  $120^\circ$ ; [1 mark]
- Joined by  $\pi$  bonds / orbitals; [1 mark]

ii) The carbon-carbon bond lengths are all the same in benzene because:

- The  $\pi$  electrons / double bonds are delocalised / in resonance

**OR**

Electrons are evenly distributed / spread out; [1 mark]

**[Total: 3 marks]**

- Each carbon forms three  $\sigma$  bonds and one  $\pi$
- The three bonds are formed from the  $sp^2$  hybridised atomic orbitals and they are at  $120^\circ$  to each other
- The  $\pi$  bonds are formed from the overlapping of the p orbitals
- A planar structure gives the maximum overlapping between p orbitals
- Evidence has shown that the carbon-carbon bond lengths are the same and the bond length is between the length of a C-C single bond and a C=C double bond
  - This information was one of the pieces of evidence that led scientists to deduce the delocalised structure of benzene

3b

b) The type of mechanism of this nitration reaction is:

- Electrophilic substitution; [1 mark]
- (Because)  $NO_2^+$  acts as an electrophile; [1 mark]
- A nitro group /  $-NO_2$  substitutes a hydrogen atom (on the benzene); [1 mark]



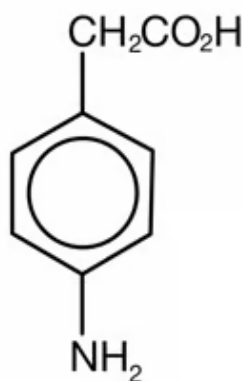
**[Total: 3 marks]**

- Benzene is refluxed with the concentrated acids at 25–60 °C
- The mixture of concentrated acids produces the electrophile  $\text{NO}_2^+$  which attacks the benzene ring
- A pair of electrons from the benzene ring is donated to  $\text{NO}_2^+$ , forming a covalent bond
- This means the benzene ring now has only four  $\pi$  bonds so there is a positive charge spread over 5 carbons, as shown by the intermediate in the mechanism
- The aromaticity is restored as the C–H bond breaks heterolytically and both electrons from this bond go into the  $\pi$  bonding system

3c

c)

i) The structural formula of (4–aminophenyl)ethanoic acid is:



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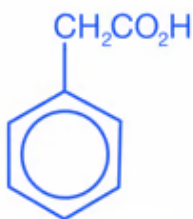
; [1 mark]

ii) The IUPAC name of the molecule shown in Fig. 3.2 is:

- 4–nitromethylbenzene; [1 mark]

**[Total: 2 marks]**

- Draw the ...phenyl)ethanoic acid part of the molecule first
- This part of the name tells you that ethanoic acid is attached to a benzene ring
- The benzene ring can only go on one carbon as the other is bonded to the carboxylic acid group:



- Now add in the 4-amino group taking the carbon on the benzene ring that is attached to the ethanoic acid as the carbon in position 1



- To name the molecule in part (ii), you should identify that the molecule is based on methylbenzene with a nitro substitute
- The carbon in position 1 is the carbon attached to the methyl group, so the nitro group is on carbon 4

3d

d) The **name** of **three** functional groups in the noradrenaline molecule are:

- Phenol; [1 mark]
- (Secondary) alcohol; [1 mark]
- (Primary) amine; [1 mark]

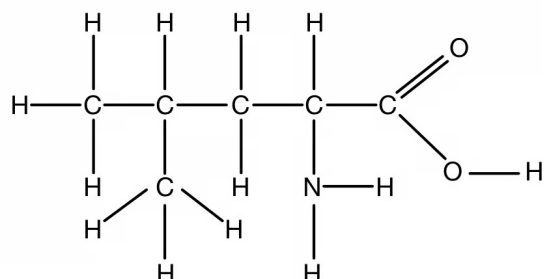
**[Total: 3 marks]**

- The -OH group is the alcohol functional group
- All the alcohol groups are attached to a carbon atom which is attached to two other carbon atoms, so it is a secondary alcohol
- The phenol functional group is a benzene ring with a hydroxyl group, -OH, attached to it
- The -NH<sub>2</sub> group is a primary amine as the N of the amine group is bonded to one R group (and two hydrogen atoms)

4a

a) The displayed formula of leucine is:

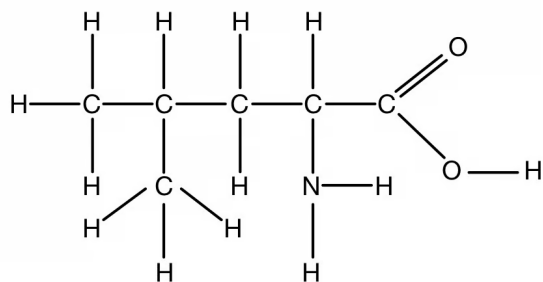
- 



; [1 mark]

**[Total: 1 mark]**

- Optical isomer questions often involve naming or drawing chemicals so you



; [1 mark]

**[Total: 1 mark]**

- Optical isomer questions often involve naming or drawing chemicals so you must be able to do this
- Ensure that you draw all of the bonds – students often forget to show the bond between O and H in the carboxylic acid group

4b

b)

i) The meaning of the term **stereoisomerism** is:

- Molecules with the same structural formula / structure

**OR**

Molecules with the same atoms bonded to each other; [1 mark]

- But with a different arrangement of atoms in space; [1 mark]

ii) You can distinguish between the isomers:

- With plane polarised light / a polarimeter; [1 mark]
- Each optical isomer / enantiomer rotates the light in opposite directions; [1 mark]

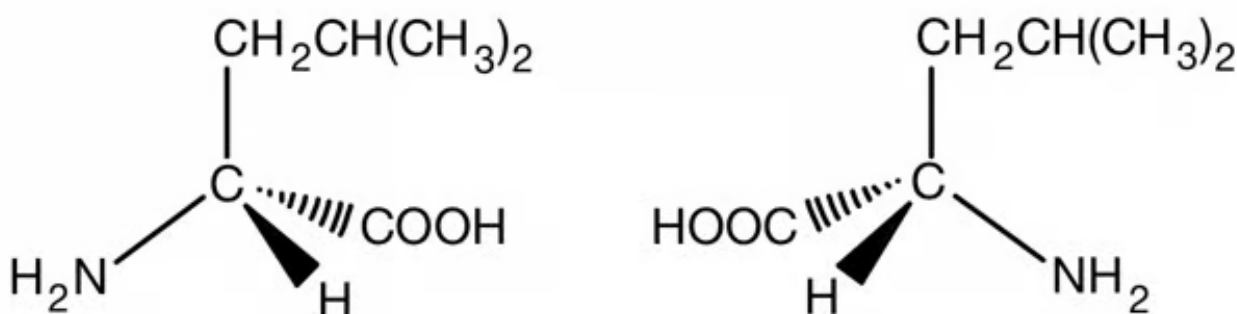
**[Total: 4 marks]**

- Both geometric and optical isomerism are examples of stereoisomerism
  - You must learn this definition word for word - it is a common definition to be asked
- Optical isomers are named like this because of their effect on plane polarised light

- You can distinguish between optical isomers because each isomer (enantiomer) will rotate the plane polarised light in opposite directions - one clockwise and the other anticlockwise
  - However, if you have a racemic mixture - a 50:50 mix of each optical isomer - then the effects will be cancelled out

4c

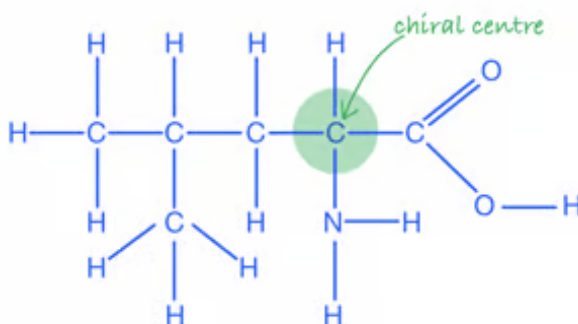
c) The three-dimensional arrangement of the two optical isomers of leucine are:



- Correct 3-dimensional structure of the left hand isomer; [1 mark]
- Correct 3-dimensional structure of the right hand isomer; [1 mark]

[Total: 2 marks]

- The functional groups can be arranged differently **but** the isomers drawn **must** be mirror images
- To determine the two optical isomers, determine where the chiral centre of leucine is:



- Draw the 3D structure with this carbon at the centre

- As the carbon has four bonds, it will be a tetrahedral shape and you must use dots and wedges to show the 3D shape
- Add the four groups attached to the chiral carbon
- Then draw the mirror image of this isomer

4d

d) The IUPAC name of leucine is:

- 2-amino-4-methylpentanoic acid; [1 mark]

**[Total: 1 mark]**

- Determine the primary chain: 5 carbons with only single bonds
  - Stem of name pentan-
- Determine the groups attached to the chain: methyl -  $\text{CH}_3$ , amino,  $-\text{NH}_2$  and carboxylic acid,  $-\text{COOH}$ 
  - Prefixes of name, amino- and methyl-
  - Suffix of name, -oic acid
- The carboxylic acid can only go on an end carbon so it does not need to be numbered but the position of the amino group and methyl group needs to be given in relation to the carboxylic acid
  - The carbon of the  $-\text{COOH}$  is given position 1, so the  $-\text{CH}_3$  group is on carbon 4 and the  $-\text{NH}_2$  group is on carbon 2
- Assemble the name, the prefixes should be listed alphabetically:
  - This gives the name 2-amino-4-methylpentanoic acid

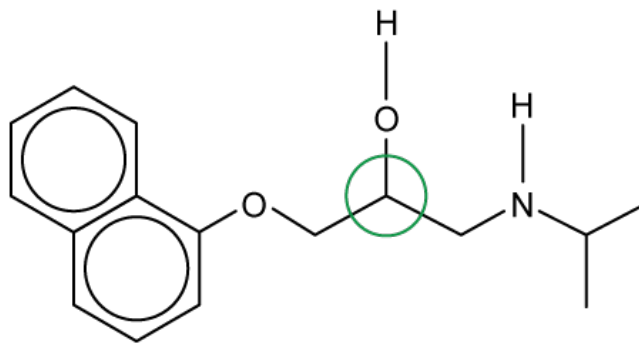
5a

a)

i) The molecular formula of propranolol is:

- $\text{C}_{16}\text{H}_{21}\text{NO}_2$ ; [1 mark]

ii) The chiral centre of propranolol is:



- Correct carbon circled; [1 mark]

**[Total: 2 marks]**

- It is very easy to miscount C and H atoms from a skeletal formula so it can help to annotate the diagram
- You could number each carbon atom - remember that the two benzene rings share two carbon atoms:

carbon 4 and the  $\text{-NH}_2$  group is on carbon 2

- Assemble the name, the prefixes should be listed alphabetically:
  - This gives the name 2-amino-4-methylpentanoic acid

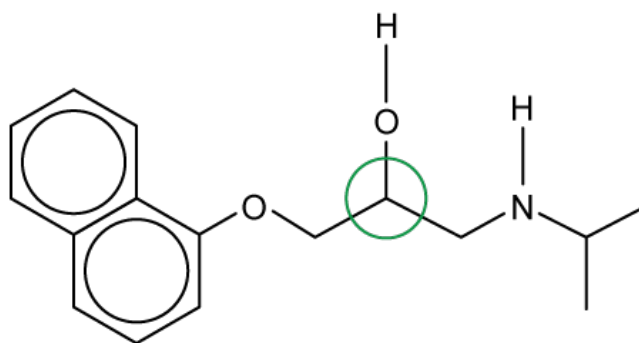
5a

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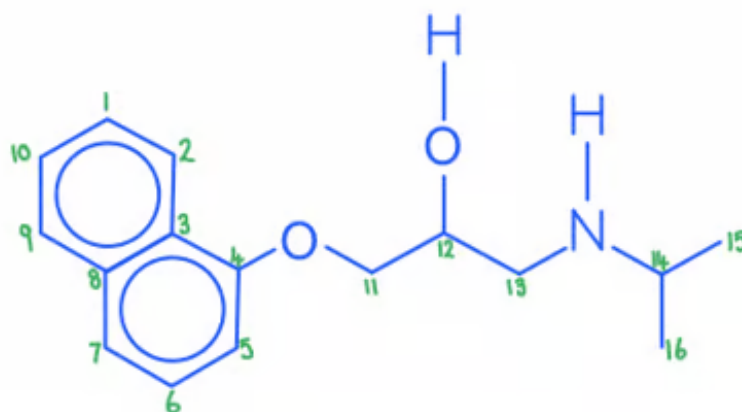


- Correct carbon circled; [1 mark]

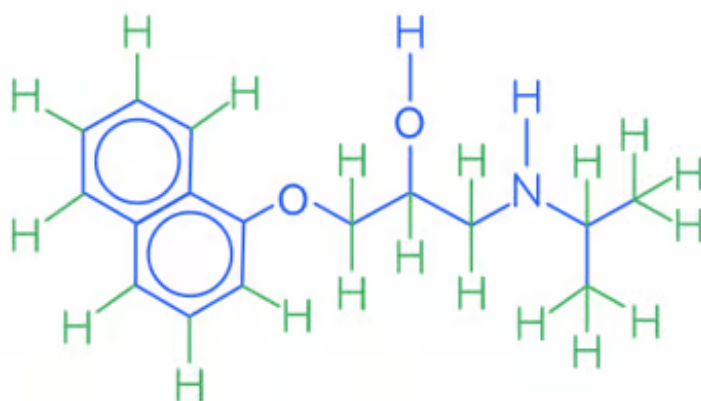
**[Total: 2 marks]**

- It is very easy to miscount C and H atoms from a skeletal formula so it can help to annotate the diagram
- You could number each carbon atom - remember that the two benzene rings share two carbon atoms:





- You could also add in the H atoms to make them easier to count:



- When writing the molecular formula of organic compounds, the convention is to write carbon, then hydrogen followed by any other elements in alphabetical order
- However, you would still be given credit if you wrote the molecular formula as  $C_{16}H_{21}O_2N$
- The carbon attached to the -OH group is the only carbon that is attached to four different atoms/groups, therefore this is the chiral centre
- Again, this can be easier to see if you draw in the hidden hydrogen atoms

5b

b) A racemic mixture of propranolol has no effect on plane polarised light because:

- Each enantiomer / optical isomer rotates plane polarised light in opposite directions

**OR**

One enantiomer rotates plane polarised light clockwise and the other anticlockwise; [1 mark]

- A racemic mixture contains equal amounts of each enantiomer / optical isomer; [1 mark]
- The effects are cancelled out; [1 mark]

**[Total: 3 marks]**

- It is important that you explain what a racemic mixture is in your response
- A racemic is said to be optically inactive
- If the mixture contains unequal amount of either enantiomer, the mixture will have an effect on plane polarised light and the mixture will be optically active

5c

c) It is important to separate a racemic mixture of propranolol into pure single enantiomers because:

- Each enantiomer has / may have different / opposite biological activity; [1 mark]
- One could have unwanted side effects

**OR**

To reduce side-effects

**OR**

To increase potency; [1 mark]

**[Total: 2 marks]**

- Enantiomers have the same chemical and physical properties, but they rotate plane polarised light in opposite directions
- However, they may have different biological activities which arise from the differences in their shape
- Each enantiomer of the drug can have quite different effects, so it is important to isolate the enantiomer which has the beneficial effects

5d

d)

i) The purpose of a chiral catalyst is:

- Ensures only one enantiomer is formed; [1 mark]

ii) **One** advantage and **one** disadvantage of using a chiral catalyst are:

- Advantage: Only small amounts of them are needed

**OR**

They can be reused; [1 mark]

- Disadvantage: Expensive; [1 mark]

**[Total: 3 marks]**

- The advantages and disadvantages of a chiral catalyst are similar to that of any catalyst
- You do need to be aware of its specific function in ensuring only one enantiomer is produced in a reaction
- Enzymes are used as biological chiral catalysts and whilst their use in drug synthesis can result in a longer process compared with the use of a more synthetic route, there are benefits in terms of sustainability and offer a 'greener'